

ICT Classified -Chapter (4) - Networks

1	Alfonso's family has three computers. He wants each member of his family to be able to access internet shopping. (a) Identify three items of hardware he may need to purchase in order to set up a computer network that will enable this to happen. 1. 2. 3. [3] Q11 – Page10 0417/11 May/June 2015
2	Aftab wants to create a network using a PC and two laptop computers he has bought for his family. All have network cards fitted. He has already purchased network protection software. He wishes to use them for online banking. Name and describe the use of four additional items, apart from cables, he would need in order to set up this network. Item 1 Use Item 2 Use Item 3 Use Item 4 Use [8] Q11 – Page10 0417/11 May/June 2015

Many computer networks are connected using cables but some are now connected using wireless technology.

..... [6]

Q15 – Page13 0417/13 May/June 2015

Describe the details of computer addresses that are stored by a router.

[illegible]

Q8 – Page5 0417/11 May/June 2016

5

Tick whether the following statements would most appropriately apply to the **internet** or an **intranet**.

	internet (✓)	intranet (✓)
A public computer network.		
Is easier to protect sensitive data from hackers.		
The content is more easily controlled.		
Exists within a single organisation only.		

[2]

Q3 – Page2 0417/12 May/June 2016

6

A school's Science department has a network which uses WiFi. A teacher has a laptop computer and whilst in school wishes to connect to this network.

(a) Write down the name of this type of network.

..... [1]

(b) In order to access the network the laptop must be within range of a type of WiFi network device.

(i) Write down the name of this type of device.

..... [1]

(ii) Describe how this device is connected to the network using an ethernet cable.

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 [1]

(c) WiFi networks can be limited in their range.

Describe one other disadvantage of a WiFi network compared to a cabled network.

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 [1]

Q6 – Page4 0417/12 May/June 2016

7

(a) Explain what is meant by VOIP.

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..... [2]

(b) Name **two** items of computer hardware that are needed in order to make a VOIP phone call from one computer to another.

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[2]

Q7 – Page4 0417/12 May/June 2016

8

Tick whether the following are features of **emails** or traditional **faxes**.

	email (✓)	fax (✓)
Transmission of messages is relatively slow.		
When received data can be edited immediately using a word processor.		
Can have attached sound files.		
The message is basically an image.		

[2]

Q2 – Page2 0417/13 May/June 2016

9

A company is setting up a video-conference.

(a) Name **three** computer hardware devices they would need.

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[3]

(b) The company could have set up a web-conference rather than a video-conference.

Describe what is meant by a web-conference.

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[4]

Q8 – Page4 0417/11 May/June 2017

Tick whether the following statements are true or false.

	true (✓)	false (✓)
A router is required to allow a LAN to connect to the internet		
A web browser displays a web page		
An intranet has restricted access		
ISP means Internal Service Provider		

[2]

Q2 – Page2 0417/12 May/June 2017

Bluetooth and WiFi can connect devices together.

Discuss the advantages and disadvantages of using Bluetooth or WiFi.

[illegible]

[8]

Q16 – Page14 0417/12 May/June 2017

12	A media company with branches around the world has recently opened a new branch in London. The branch includes a number of different computer networks. Identify the most appropriate type of network to answer each of the questions. (a) The type of network used to connect the branch to its head office in New York is called a [1] (b) The type of cabled network used to connect computers together in one of the offices is called a [1] Q3 – Page2 0417/11 May/June 2018
13	A motor car company has some designers based in London and some in Beijing. The cost of travel between the two cities is very high, so when they wish to meet to discuss new products they use video-conferencing. (a) The designers all have PCs with a keyboard and a mouse in order to take part in video-conferencing. Name three other devices used to input or output data which would be needed to take part in the video-conference. 1 2 3 [3] (b) Describe three potential problems of the designers using video-conferencing systems rather than meeting in either London or Beijing. 1 2 3 [3]

(c) Documents that are sent between the two cities are encrypted.

Explain what encryption is and why it is used.

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[3]

Q12 – Page10 0417/11 May/June 2018

14

Tick the most appropriate type of network for each description.

	LAN (✓)	WAN (✓)	WLAN (✓)
A cabled network in a building.			
A network that can use satellites.			
A network that connects countries.			
A network that uses wireless technology.			

[4]

Q3 – Page2 0417/12 May/June 2018

15	A company used to have video-conferences with its suppliers, but due to technical problems they are considering running an audio-conference. Describe the advantages of using audio-conferencing rather than video-conferencing. [4] Q12 – Page13 0417/12 May/June 2018
16	An organisation is planning to train all staff online rather than by video-conference. (a) Describe what is meant by a web-conference. [3]

(b) Explain the differences between video- and web-conferencing.

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[3]

Q13 – Page12 0417/13 May/June 2018

17

The manager of a large company wants to replace the company's cabled LAN with a WLAN.
Describe the features of a WLAN.

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[6]

Q16 – Page13 0417/13 May/June 2018

18

A number of devices are used to enable computers to communicate within a network. Two of these devices are hubs and switches.

(a) Compare and contrast the use of a hub and a switch.

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[4]

(b) Bluetooth is an alternative method of allowing computers to communicate with other computers and devices.

Describe how a tablet computer can connect to a printer using Bluetooth.

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[4]

Q7 – Page5 0417/13 May/June 2019

Sometimes specialist books are needed in the library. When this happens a message has to be sent to the main library in the town. The message can either be sent by email or by physical fax.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[6]

Q11 – Page12 – b 0417/13 May/June 2019

20

Tick (✓) which of the following statements apply to LANs, which apply to WANs and which apply to WLANs.

	LAN (✓)	WAN (✓)	WLAN (✓)
The internet is an example of this type of network			
This type of network is found in a building but is connected without cables			
This type of network is used to transmit data between Europe and North America			
This type of network is found in a building but is connected with cables			

[4]

Q6 – Page8 0417/11 May/June 2020

21

Complete the following sentences using the most appropriate network term.

A communication method that wirelessly connects a mobile phone to a car, to allow hands-free use of the mobile phone is

A network device that allows a stand-alone computer, using a dial-up analogue connection, to connect to an ISP is a

A network device that allows a LAN to connect to the internet is a

A wireless communication method used in WLAN is

[4]

Q3 – Page5 0417/12 May/June 2020

[6]

[2]

(b) Explain what a WAN is.

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..... [2]

Q3 – Page3 0417/13 May/June 2020